

Code smell: Long method - "getAvailableDirections" is too long in the monster class

Resolution: extracted parts of the main method into 2 smaller ones

Result: cleaner looking code and increased readability

Code smell: Dead code – public getter methods are never used

Resolution: deleted the obsolete methods

Result: reduced code size

Code smell: Long method - "getPath" is too long in the pathfinder class

Resolution: split getPath into 2 smaller methods

Result: cleaner looking code, easy readability, can be easily modified

Code smell: Repeated code – repeated if block in the monster class to check collision of the player

Resolution: extract the if block into its own method called collisionCheck

Result: cleaner looking code and can be easily modified if needed

Code smell: Repeated code – loops repeated to instantiate walls the map class

Resolution: extract the repeated for loops into methods for creating vertical and horizontal walls

Result: cleaner looking code, reduced method size, easily modified methods

Code smell: Long method – after refactoring repeated code in map class, method is still too long

Resolution: create 2 methods, one for creating the vertical walls, and another for the horizontal walls

Result: decreased the main method size

Code smell: Performance concern – monster class instantiates a pathfinder object every time it wants to create a path to the player. Pathfinder is a bulky class, instantiation of it is expensive

Resolution: create a singleton of the pathfinder in the round object and change the methods so that it mutates its own existing fields

Result: increase in performance

Code smell: Poorly named variable – Pathfinder class has a const variable named DX but it doesn't really describe what it is for

Resolution: change name to INCREMENT

Result: provides more readable code, and less obscurity